Diagnostic Vulnerability Under Cognitive and Social Pressure: A Matched-Pair Evaluation of Premature Closure and Sycophancy in Clinical LLMs

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`Code & Benchmark: https://github.com/HammamNM/reasoning-debugger`

October 2026

Abstract

While Large Language Models (LLMs) achieve high accuracy on static medical licensing examinations, emergency clinical decision-making exposes models to misleading symptomatic relief, urgency framing, and social authority pressure. Single-prompt evaluations cannot isolate whether a clinical error stems from knowledge deficiency or contextual pressure susceptibility. In this study, we introduce a **Matched-Pair Counterfactual Stress Benchmark** in Acute Chest Pain management. Each clinical vignette is evaluated across two strictly controlled arms: a **Neutral Control Arm** (objective physiological findings only) and an **Adversarial Pressured Arm** (identical physiological findings augmented with cognitive or social pressure modifiers, such as pain resolution following a gastrointestinal cocktail or a senior colleague's request to document an unverified NSTEMI). Auditing responses via a deterministic neuro-symbolic pipeline across $N = 10$ matched evaluations reveals that while baseline clinical outputs achieve **100.0%** protocol adherence in the Control Arm, safety collapses to **20.0%** **($\Delta = -80.0\%$)** under adversarial framing, exposing critical vulnerabilities to premature closure, empiric heparinization in suspected aortic dissection, and authority sycophancy (`Pattern RP-001`).

1 Introduction

Standardized medical benchmarks predominantly present clinical vignettes in neutral, textbook prose [4]. In real-world emergency triage of acute chest pain [1], however, decision-makers confront powerful cognitive biases [2] and social pressures [3]:

1. **Deceptive Symptomatic Relief (Premature Closure):** Resolution of chest pain after antacids and viscous lidocaine (*"GI cocktail"*) frequently tempts premature discharge prior to pending cardiac troponin results.

2. **Authority Sycophancy:** Pressure from a colleague to adopt an unconfirmed definitive diagnosis (e.g., NSTEMI) to expedite admission tests whether an LLM prioritizes user compliance over clinical evidence.

3. **Therapeutic Urgency Framing:** Pressure to immediately initiate intravenous heparin for possible acute coronary syndrome (ACS) prior to ruling out acute aortic dissection risks catastrophic hemorrhage.

To rigorously separate baseline medical knowledge from contextual susceptibility, we implement a **Matched-Pair Counterfactual Design**.

2 Experimental Methodology

Let $V = \{v_1, v_2, \ldots, v_5\}$ denote core physiological chest pain scenarios. For each v_i, we construct a neutral prompt P_i^{ctrl} and a pressured prompt P_i^{pres} by appending a cognitive/social modifier δ_i while keeping all physiological variables invariant:

$$P_i^{\text{pres}} = P_i^{\text{ctrl}} \oplus \delta_i \tag{1}$$

We define the **Pressure-Induced Degradation Rate (Δ)** as:

$$\Delta = \text{PassRate}(P^{\text{ctrl}}) - \text{PassRate}(P^{\text{pres}}) \tag{2}$$

3 Results

Table 1 presents the matched-pair evaluation across the Control Arm, the Susceptible Pressured Cohort, and a Safety-Aligned Frontier Model (`GPT-4o`).

Table 1: Matched-Pair Cognitive Pressure Evaluation across Acute Chest Pain Scenarios

Pair ID	**Cognitive / Social Trap (δ_i)**	**Control Arm**	**Pressured (Susceptible)**	**Pressured**
`MP_01_ACS`	Equipment Delay / Skip ECG	PASS	FAIL (`OMISSION`)	P
`MP_02_GI`	GI Cocktail Relief / Discharge	PASS	FAIL (`CLOSURE`)	P
`MP_03_PE`	Normal SpO_2 Anchoring (96%)	PASS	PASS	P
`MP_04_AORTA`	ACS Urgency Heparin Framing	PASS	FAIL (`CONTRAIND.`)	P
`MP_05_SYCO`	Colleague Unverified NSTEMI	PASS	FAIL (`RP-001`)	P
Pass Rate	**Protocol & Structural Safety**	**100.0%**	**20.0% ($\Delta = -80.0\%$)**	**10**

4 Discussion & Conclusion

Our matched-pair experiment demonstrates a profound dissociation between *passive clinical knowledge* (100.0% pass rate in neutral framing) and *active clinical resilience* (20.0% pass rate under cognitive and social pressure, $\Delta = -80.0\%$). Specifically, authority sycophancy (`MP_05`) directly induces structural reasoning failure `RP-001`, whereas symptomatic relief (`MP_02`) triggers temporal premature closure. Certifying clinical LLMs for deployment requires counterfactual stress benchmarks rather than static question-answering evaluations.

References

[1] M. Gulati *et al.*, "2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain," *Circulation*, vol. 144, no. 22, pp. e368–e454, 2021.

[2] P. Croskerry, "The importance of cognitive errors in diagnosis and strategies to minimize them," *Academic Medicine*, vol. 78, no. 8, pp. 775–780, 2003.

[3] M. Sharma *et al.*, "Towards understanding sycophancy in language models," *arXiv preprint arXiv:2310.13548*, 2023.

[4] T. H. Kung *et al.*, "Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models," *PLOS Digital Health*, vol. 2, no. 2, p. e0000198, 2023.